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# Integration of non-terrestrial network for 5G NR and future 6G: LEO satellite-to-device performance and interference analysis

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ABSTRACT

Non-Terrestrial Networks (NTN) are considered pivotal for the development of 6G, aiming to provide ubiquitous and continuous mobile broadband coverage. With ongoing standardisation efforts by 3GPP, 5G NTN promises seamless connection moving between terrestrial and satellite networks, using existing or next-generation smartphone devices. This paper focuses on the integration of NTN, particularly Low Earth Orbit (LEO) constellations, for 5G NR services. We explore the latest developments in direct satellite-to-device communication and the significant challenge posed by interference. We assess the performance of 5G NTN by simulating and analysing the downlink performance of LEO constellations providing continuous service to moving User Equipment (UE) in unserved and remote areas, addressing the challenges of achieving higher data rates and managing signaling interference. Based on our baseline Link Budget calculation result, higher Effective Isotropic Radiated Power (EIRP) and additional spectrum bandwidth are essential for enhancing data rates and user coverage. However, given the limited availability of spectrum, increasing EIRP becomes a more practical option to increase data rates, which in turn increases the risk of interference. Our simulation findings indicate that LEO constellations can provide continuous coverage for moving UEs, supporting low data rate services such as text, voice and Internet of Things (IoT), but the limited spectrum bandwidth and interference proved to be some of the main challenges for achieving higher data rates. Interference management strategies, such as reducing sidelobe power and employing advanced technologies such as beamforming [1] and beam-hopping [2], are critical to mitigate interference and improve performance.

## 1. Introduction

Non-Terrestrial Network (NTN) is considered one of the main ingredients to achieve 6G's envision of providing true global coverage with faster data rate and higher capacity than 5G. The standardisation of integrating NTN with 5G-NR and NB-IoT in 3GPP provides incentives for smartphone chip vendors, Mobile Satellite Service (MSS) providers, Satellite Operators, and Mobile Network Operators (MNOs) to join forces in incorporating the standards into their products and networks. This standardisation eliminates the current need for separate proprietary terminals for direct satellite-to-Device (D2D) service, significantly broadening the use case and making the 6G vision of connectivity anywhere, anytime possible.

Various countries, such as Australia, are replanning the use of the proposed NTN bands defined by 3GPP in Rel 17 and Rel 18 and moving toward the commercial launch of D2D and Satellite IoT services [3]. While a series of collaborative trial tests between relevant parties are in full swing, concerns over harmful interference have emerged. Recent complaints about potential interference between satellite operators due to the overlapping of mobile and the 3GPP NTN bands drawn investigations from industry and government agencies (Omnispace case toward T-mobile/Starlink [4], National Academies of Sciences, Engineering, and Medicine on Ligado Networks [5]). Such concerns underscore the need to study interference effects and adopt appropriate regulatory or technological solutions before allocating additional spectrum.

In this study, we analyse the integration of NTN, particularly the LEO constellation, for 5G NR services. We delve into the latest developments concerning direct satellite-to-device communication, its performance, and the significant challenge posed by interference. According to Euroconsult's, direct satellite-to-device services could potentially connect nearly 130 million average monthly users by 2032 [6], supported by the latest development of partnerships between MNOs, MSS providers and Satellite operators leveraging current mobile spectrum and smartphones, or utilising assigned MSS and 3GPP NTN bands for direct-to-device services. While recent services focus on low-data-rate communications, the release of Rel-18 5G NR-NTN introduces support for new

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frequency bands above 10 GHz in the Ka band, targeting devices with larger antennas, such as Very Small Aperture Terminals (VSATs), for high-speed broadband connections.

Offering seamless and continuous worldwide coverage are the key envision for 6G. Among the different satellite systems (LEO, MEO, GEO), LEO system systems offer reliable, low-latency, and high-speed Internet from space, extending cellular 5G networks to air, sea, and remote areas not covered by Terrestrial Networks (TNs). This is particularly valuable for countries like Australia, Canada and United State, with vast landmass and sparsely populated regions prone to natural disasters, such as bushfires and flooding, where maintaining continuous and reliable communication is essential for coordinating rescue and relief efforts. As the launch of new LEO satellites and the formation of partnerships between MNOs, MSS providers and Satellite operators continue to accelerate, the performance of LEO-based 5G NTN systems, especially for enhanced Machine Type Communication (eMTC) and enhanced Mobile Broadband (eMBB), is of significant interest. However, large-scale LEO constellations for 5G coverage present challenges, notably signal interference between satellite systems.

In this paper, we simulate and analyse the downlink performance of LEO constellations, evaluating how well they can maintain service continuity for moving User Equipment (UE) in unserved and remote areas. We delve into the latest developments concerning direct D2D communication and review the recent research on NTN 5G NR performance analysis, interference effect and management. We also consider how co-channel interference among satellite systems affects key metrics such as Carrier-to-Noise Ratio (CNR), Energy per Bit to Noise Power Spectral Density (Eb/N0), and throughput. In the end, we offer analytical insights into the performance results and discuss current constraints and future work directions on interference management.

The contribution of this work:

- In this paper, we provided an update on NTN's latest development, together with a technical overview of 5G-NR NTN using LEO constellation. The corresponding interference challenge is also discussed.
- The examination is based on simulations of a real-world scenario in terms of providing continuous coverage, achievable bit rate and throughput capacity.
- We explored the impact of interference on Quality of Service (QoS) for handheld devices with low G/T ratios in D2D links. While previous studies have primarily focused on interference at ground stations and between satellites, our work fills a critical gap in current research on direct satellite-to-device communication.
- We examined the interference effect between different LEO constellation systems on the downlink to the user and evaluated the extend of service degradation based on CINR and throughput loss. We quantify the trade off between EIRP, sidelobe and CINR across LEO constellations.
- We demonstrate that increasing EIRP without adequate sidelobe power reduction can cause severe degradation for neighbouring or competing satellite constellation systems, underscoring the need to establish operational boundaries for sidelobe transmission power between systems.
- We also offered an analytical framework using simulation that can facilitate future NTN system design and further expand onto use cases such as satellite to moving vehicle/vessel in this paper.

The structure of the paper is organised as follows: Section 2 will provide an update on NTN development, including a technical overview of LEO constellations, 5G-NR, and interference. Section 3 will present Literature review on 5G-NR NTN performance and the interference challenge. We will illustrate the system model in Section 4 and Section 5. The result and discussion will be presented in Sections 6 and 7. Section 8 will cover future work, and the paper will be concluded in Section 9.

**Table 1**
3GPP NTN bands.

| 3GPP NTN bands | Band# (Duplex) | Uplink | Downlink |
|---|---|---|---|
| 3GPP FR1 (L&S Band) | n256 (FDD) | 1980–2010 MHz | 2170–2200 MHz |
| | n255 (FDD) | 1626.5–1660.5 MHz | 1525–1559 MHz |
| | n254 (FDD) | 1610–1626.5 MHz | 2483.5–2500 MHz |
| 3GPP FR2 (Ka Band) | n510 (FDD) | 27.5–28.35 GHz | 17.7–20.2 GHz |
| | n511 (FDD) | 28.35–30 GHz | 17.7–20.2 GHz |
| | n512 (FDD) | 27.5–28.35 GHz | 17.7–20.2 GHz |

**Table 2**
NTN satellite characteristic in different access network scenarios.

| | IoT NTN Narrowband | NR NTN Narrow/Wideband | NR NTN Broadband |
|---|---|---|---|
| Frequency band | L-band, S-band | L-band, S-band | Ka-band |
| Radio Interface | NB-IoT | 5G NR | 5G NR |
| Service bit rate | Up to 100 + kbps | Up to few Mbps | Up to 100 + Mbps |
| Terminal | IoT device | IoT / Smartphone | VSAT |
| Use case | Data - e.g. Smart grid, utilities | Data and voice e.g. Consumers, public safety | Data,voice,video e.g. Backhaul, IPTV |

**Table 3**
Proposed spectrum allocation for NTN NR and IoT in Australia.

| Bandwidth | Uplink | Downlink | Allocation approach | Service |
|---|---|---|---|---|
| 2x 25 MHz | 1980–2005 MHz | 2170–2195 MHz | Auction | Direct to UE/IoT |
| 2x 5 MHz | 2005–2010 MHz | 2195–2200 MHz | Shared | Direct to IoT |

## 2. 5G NTN development update and technical overview of LEO constellation, 5GNR and interference challenges

### 2.1. NTN integration development update

In 3GPP Release 17 and the latest Release 18, spectrum allocation for IoT and NR NTN in the L-band and S-band, as well as the Ka band for higher bit rate services, has been specified. Release 18 of 5G NR-NTN includes additional allocation in the L-band and S-band, increasing the allocation for frequencies below 7 GHz (FR1) to 80 MHz for both Downlink (DL) and Uplink (UL) [7]. It will also support new frequency bands for NTN above 10 GHz, such as the Ka band, for much higher carrier bandwidths and network capacities. The Ka band assignment does not target smartphones but devices equipped with larger antenna or antenna arrays, such as VSAT. Table 1 illustrates the allocation in detail [7,8]. Table 2 summarises the characteristics of different access scenarios based on the latest 3GPP release [9].

In Australia, for example, 2 × 25 MHz will be replanned for satellite direct-to-device communications service. Another 2 x 5 MHz dedicated for satellite IoT and similar narrowband services, the spectrum is proposed to be shared on a basis between operators to provide a low barrier to entry [3]. Table 3 summarises the proposed allocation [3]:

While the available bandwidth may initially be insufficient to support high data rates and voice and data services directly to smartphones, early offerings of these constellations will likely be limited to emergency alerts and messaging. Satellite operators and MNOs are both keen to capitalise on this potential market. Satellite operators gain exposure to a much larger potential market than they can reach today with expensive proprietary devices [8]. MNOs are focused on integrating satellite D2D services as a competitive edge, leveraging partnerships with satellite operators to enhance the telco customer experience with ubiquitous coverage.

Based on recent developments, two strategic approaches dominate the utilisation of scarce spectrum resources for direct satellite-to-device (D2D) services based on 5G NTN. Satellite operators such as Viasat, Ligado Networks, and Omnispace advocate for utilising MSS spectrum

**Table 4**
Mobile and satellite operator partnership using MSS spectrum.

| Operators | Satellite system | Spectrum | Technology | Services |
|---|---|---|---|---|
| Omnispace + MTN Africa [19] | 200 LEO | S band | 3GPP-NTN Rel 17 | Messaging, Data, Voice, Video |
| Sateliot [14] | 250 LEO | 2 GHz | 3GPP-NTN Rel 17 | NB-IoT |
| Ligado [14] | GEO | L band | 3GPP-NTN Rel 17 | NB-IoT |
| Qualcomm + Iridium [14] | 66 LEO | Ka band | Proprietary | Messaging |
| Kuiper + Verizon [14] | 3236 LEO | Ka band | Proprietary | Backhaul LTE/5G |

**Table 5**
Mobile and satellite operator partnership using MNO spectrum.

| Operators | Satellite system | Spectrum | Technology | Services |
|---|---|---|---|---|
| Starlink + T-mobile, Optus [20,21] | 2016 LEO | MNO spectrum | 3GPP- release 12 | Data, Voice, Video |
| Starlink + KDDI [12] | 2016 LEO | MNO spectrum | 3GPP- release 12 | Backhaul |
| AST + AT&T, Rakuten [15,22] | 243 LEO | MNO spectrum | 3GPP- release 12 | Data, voice, video |
| Lynk + Roger Canada, Telstra [11,23] | 5000 LEO | MNO spectrum | Pre 3GPP-NTN release | Messaging, Low Data rate service |

already licensed to them and specific spectrum band (presented in Table 1) allocated for NTN technologies by 3GPP to connect user devices, rather than relying on spectrum derived from terrestrial MNOs [10]. Conversely, operators such as SpaceX, Lynk Global, and AST SpaceMobile aim to utilise spectrum within frequency bands designated for terrestrial MNOs [10]. In Australia, Optus announced its collaboration with Starlink in 2023 and planned to support D2D SMS messaging by the end of 2024 and voice/data in 2025. Canada's Rogers has collaborated with Lynk for D2D services, beginning with SMS messaging, mass notifications, and machine-to-machine AI applications before expanding into voice and data services [11]. In Japan, KDDI has established a partnership with Starlink to deliver services including backhaul connectivity, text messaging, voice, and data services [12]. Similarly, Switzerland's Salt network has adopted Starlink's constellation to enhance its terrestrial infrastructure, initially focusing on backhaul and backend network connections and text messaging service before expanding to voice and data services [13]. Tables 4 and 5 summarise some of the partnerships and their use of spectrum bands [14,15].

In 2023, AST SpaceMobile, in partnership with Vodafone, AT&T, and Nokia, conducted 5G direct-to-device tests using a Samsung Galaxy S22, successfully reaching data transfer rates of 14 Mbps [16]. Earlier in 2024, SpaceX/Starlink utilised T-Mobile's spectrum, connecting with multiple models of Samsung, Apple, and Google devices and delivering successful SMS and Over-the-top (OTP) messages and achieving data download speeds of 17 Mbps [17].

In Europe, the ITA-NTN project, conducted under the European Union's Italian National Recovery and Resilience Plan (NRRP) RESTART Research Program, provides a comprehensive European perspective on Terrestrial/Non-Terrestrial (T/NT) network integration for future 6G systems [18]. The research examines different T/NT network integration architectures, including drone-based relay networks, satellite-based access and backhauling, and 3D single or multi-connectivity configurations utilising LEO/GEO satellites, HAPs, and UAVs. By mapping specific use cases such as emergency operations, transport systems, maritime applications, and rural connectivity to appropriate T/NT network architectures, the project addresses Europe's unique connectivity challenges, particularly in urban/suburban areas data offloading, remote area emergency operation management and transport systems in high speed (i.e., trains and automotive) scenarios. This European framework offers actionable guidance for operators and stakeholders on how NTN can serve as RAN or backhaul to address Europe's digital divide, enhance network resilience, and support applications ranging from drones for delivery, interactive aerial telepresence with augmented reality to autonomous maritime operations.

This project complements these global NTN integration initiatives by providing quantitative analysis of the performance limitations that different NTN architectures must address. While frameworks such as the ITA-NTN project provides architectural examples for multi-layer connectivity using satellites, HAPs, and UAVs, our research fills a crucial gap by quantifying the downlink performance constraints and interference challenges that these architectures may face in practice. Additionally, while partnerships between MNOs and satellite operators (such as T-mobile-Starlink, KDDI-Starlink and Rogers-Lynk) focus on service deployment, our analysis reveals the underlying technical challenges of spectrum scarcity and dynamic interference that must be addressed in order to scale beyond basic messaging services to higher data rate applications. This performance analysis is valuable for validating whether proposed 2D/3D connectivity architectures can deliver their intended services under real-world interference conditions and spectrum constraints.

### 2.2. LEO constellation - technical overview

As defined by 3GPP, a Non-Terrestrial Network (NTN) is a communication network that partially or fully operates via airborne platforms (e.g., UAVs, high-altitude platforms) or space-borne vehicles (including

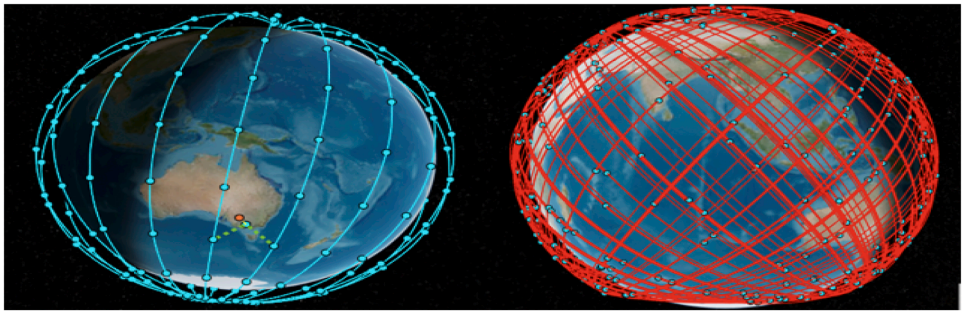


**Fig. 1.** Walker star and walker delta constellation.

GEO, MEO, and LEO satellites). NTN systems address coverage gaps in terrestrial networks (TN) and enhance network performance by providing a unique vantage point for extending connectivity to remote and underserved regions. Among the GEO, MEO and LEO, LEO satellites offer latency as low as tens of milliseconds and provide capacity in the range of tens of megabits per second. These attributes make LEO satellites well-suited for Mobile Broadband (MBB) and Machine-Type Communication (MTC) applications, aligning with the evolving requirements of both 5G and 6G technologies. In this section, we will delve into the detailed characteristics of LEO constellations.

One of the distinctive features of LEO satellites is their smaller footprint, which is the coverage area served by the spot beams, compared to MEO and GEO satellites. To ensure continuous coverage despite this limited footprint, LEO constellations typically consist of larger numbers of satellites. Satellite operators such as Starlink, OneWeb, and Amazon Kuiper have been the market leaders in deploying these large LEO constellations, primarily targeting fixed broadband internet services for residential and business users.

LEO constellations generally utilise circular orbits and can be classified into two main categories based on their orbital characteristics:

- Walker Star or Polar Constellations: These constellations use near-polar inclinations, distributing the ascending nodes of the orbits across a hemisphere 180°, with higher inclinations, this design can cover both poles and lower latitudes [24]. Examples include the Iridium and OneWeb constellations.
- Walker Delta Constellations: These constellations distribute ascending nodes uniformly across 360°of longitude, which ensures that any area on Earth has both ascending and descending satellites, making it suitable for mid-latitudes [24]. Starlink and Globalstar are examples of constellations employing this design. Fig. 1 shows the two constellations using a MATLAB simulation.

LEO satellites travel at velocities ranging from 7.2 to 7.5 km/s relative to a fixed point on Earth, resulting in orbital periods between 90 and 110 m, where the LEO constellation altitude is around 300–700 km [25]. This rapid movement means that a single satellite communicates with a ground station/user for approximately 5 to 15 m over 6 to 8 passes per day [25]. Communication links between LEO satellites and ground stations or UE are established during satellite passes when the satellite is within visibility range [24]. Due to the movement of LEO satellites, the distance between the satellite and the ground station/UE varies over time, impacting the signal's path loss and consequently affecting the Carrier-to-Noise Ratio (CNR) and the performance of receiving ground stations/devices [26].

Latest LEO satellites utilise multi-beam antennas and phased arrays to generate numerous high-gain narrow beams [27]. This configuration boosts the system's capacity throughout its coverage area [27], as each beam's gain increases with a decrease in antenna beamwidth [28]. Direct-radiating arrays are also employed due to their wide scanning angles and superior off-boresight performance compared to phased array antennas [27]. AST SpaceMobile has launched the largest antenna for LEO satellites, featuring a 64-square-meter phased array. Starlink's 2nd Gen satellites, expected to support direct-to-device capabilities, are equipped with 24-square-meter phased array antennas. These antennas can dynamically steer multiple beams across the Earth's surface, supporting a total of 48 downlink beams and 16 uplink beams [29]. In addition, frequency reuse is achieved by alternating frequency sub-bands between beams, allowing for increased system capacity without expanding the bandwidth [28].

### 2.3. 5G NR - technical overview

Similar to LTE, 5G NR adopts the Orthogonal Frequency Division Multiplexing (OFDM) transmission scheme as its default waveform for both downlink and uplink directions. One of the distinct differences from LTE is that 5G NR supports flexible OFDM numerology, allowing for the customisation of sub-carrier spacing (SCS) and cyclic prefix (CP) length [30]. This results in a more scalable and flexible physical layer compared to LTE. The use of sub-carrier spacing ranging from 15 kHz up to 240 kHz enables 5G NR to support higher frequencies and a wider range of deployments, from large cell to mmWave deployments with wide spectrum allocations [30]. Table 6 highlights some of the physical parameters of 5G NR.

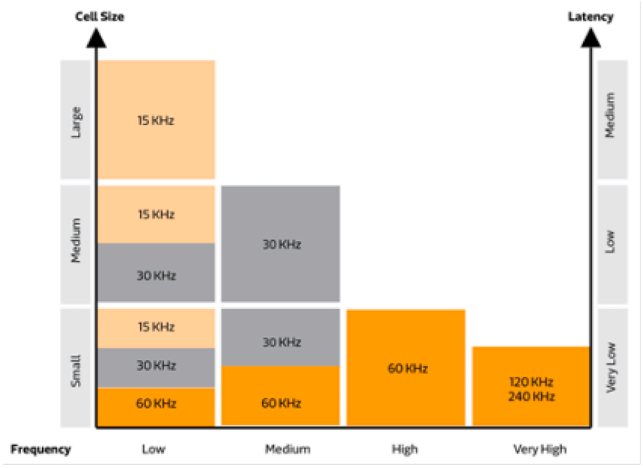



**Fig. 2.** The relationship between numerologies, frequency and latency [31].

The selection of OFDM sub-carrier spacing is determined based on factors such as carrier frequency, phase noise, delay spread, and Doppler spread, with different numerologies tailored to diverse deployment scenarios and their corresponding performance requirements [32]. Fig. 2 depicts the relationships between SCS, frequency, cell size and latency, as illustrated, smaller sub-carrier spacing facilitates larger cell sizes, making it suitable for lower frequency deployments, while larger SCS enhance latency performance due to shorter symbol duration [31]. However, smaller SCS may lead to performance degradation in high Doppler scenarios, indicating a trade-off between system robustness and spectrum efficiency.

In terms of bandwidth, 5G NR supports up to 100 MHz bandwidth in lower frequency (FR1) and 400 MHz in higher frequency (FR2), with carrier bandwidth limited to 400 MHz but larger bandwidths supported through carrier aggregation. Given the much wider bandwidth 5G NR supports, it introduced the receiver-bandwidth adaptation using Band-Width Part (BWP), allowing UEs to use narrower bandwidths for monitoring control channels, full bandwidth can be opened dynamically when higher data rate is needed, thus providing flexibility to reduce bandwidth and power consumption for UEs [33].

The time domain structure of 5G NR comprises frames and subframes, with each subframe divided into slots containing 14 OFDM symbols [33]. In the frequency domain, resource elements consist of one subcarrier during one OFDM symbol, and 12 consecutive subcarriers form a resource block; a minimum of 20 resource blocks is necessary for the UE to synchronise with the carrier [30]. 5G NR's physical resource block (PRB) is used for actual transmission with UEs, with each UE is capable of occupying one or more PRBs. Fig. 3 indicates the 5G-NR channel and bandwidth from 3GPP Rel 15. Table 7 summarises the numerologies, resource block and bandwidth information and shows an example of the number of supported PRBs for 5 MHz channel bandwidth.

### 2.4. NTN current interference challenges

The rapid expansion of mega-constellations, comprising thousands of satellites operating across various altitudes and orbital planes, poses significant challenges for spectrum management. Limited available spectrum, frequency reuse, and the low antenna gain of handheld/IoT devices increase the risk of harmful interference, complicating the

**Table 6**
5G - NR physical layer specifications.

| Parameters | UPLINK | DOWNLINK |
|---|---|---|
| Transmission scheme | OFDM | OFDM |
| Multiple Access scheme | OFDMA | OFDMA |
| Frequency range | FR1 - 450 MHz to 6 GHz<br>FR2 – (mmWave) - 24.25 GHz to 52.6 GHz | |
| Bandwidth | 5-100MHz (FR1) 50-400MHz (FR2) | 5-100MHz (FR1) 50-400MHz (FR2) |
| Subcarrier spacing (SCS) | 15, 30, 60, 120, and 240 kHz | 15, 30, 60, 120, and 240 kHz |
| Modulation | BPSK, QPSK, 16QAM, 64QAM, 256QAM | QPSK, 16QAM, 64QAM, 256QAM |
| Duplex schemes | FDD/TDD for FR1<br>TDD for FR2 | FDD/TDD for FR1<br>TDD for FR2 |

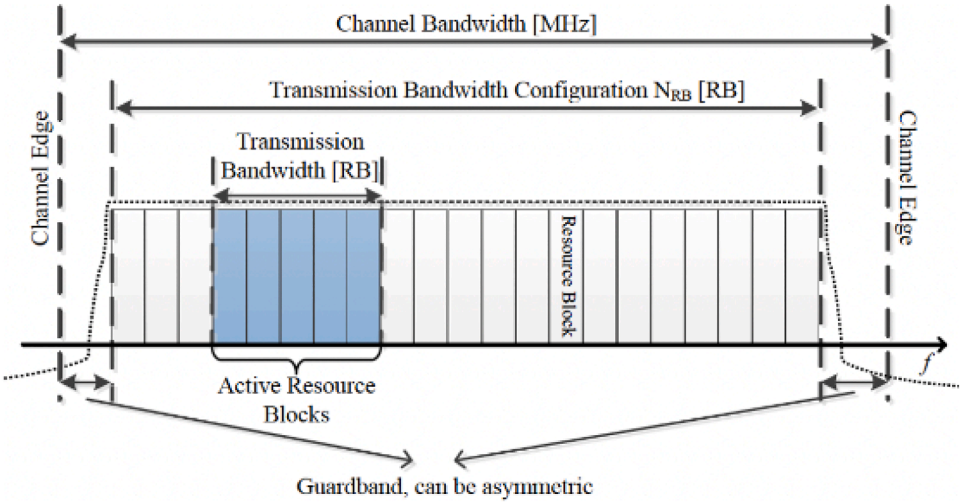



**Fig. 3.** Channel and bandwidth for one NR channel [34].

**Table 7**
5G NR- numerology ($\mu$) for different frequency ranges.

| $\mu$ | Freq. range | SCS (kHz) | PRB size (kHz) | Min BW (MHz) | Max BW (MHz) | Number of PRBs for 5 MHz BW |
|---|---|---|---|---|---|---|
| 0 | FR-1 | 15 | 180 | 4.32 | 49.5 | 25 |
| 1 | FR-1 | 30 | 360 | 8.64 | 99 | 11 (N/A) |
| 2 | FR-1 /FR-2 | 60 | 720 | 17.28 | 198 | N/A |
| 3 | FR-2 | 120 | 1440 | 34.56 | 396 | N/A |
| 4 | FR-2 | 240* | N/A | 69.12 | 397.44 | N/A |

integration of NTN. In satellite systems, Radio Frequency Interference (RFI) can occur in various forms [27], generally defined as:

- Co-channel Interference: This happens when the signal quality of the user receiver is impacted by multiple satellites or ground stations operating on the same frequency.
- Adjacent Satellite Interference: This occurs when the user terminal or satellite is affected by interference operating on a nearby frequency, causing slight overlaps.

RFI can originate from several sources, including [28]:

- Between satellites within the same system.
- Between different satellite systems.
- Between satellite systems and terrestrial communication systems.
- Within the same satellite due to different beams (e.g., self-interference).

Interference is a concern both on the uplink and downlink paths. In this study we will focus on the service link from the satellite to ground users using handheld/IoT devices, as transmitter power and receiver gain are restricted compared to the feeder link. We are particularly interested in inter-satellite system interference where there is no direct coordination or communication between systems.

#### 2.4.1. Interference dynamics and impact

One of the major differences between current fixed or mobile satellite service and the NTN satellite direct-to-device service is the low-gain antennas of handheld / IoT devices. Geostationary satellite systems (GSO) can share frequency bands by leveraging the fixed geometries and directional antennas of ground stations/VSAT [26]. The situation becomes more complex when the LEO satellites appear to be constantly moving and when the receiving handsets with low-directivity antennas are used. For NTN's D2D system, the handheld UE is typically equipped with omnidirectional antennas, with a gain-to-noise ratio (G/T) of -31 dB/K [35]. To ensure connectivity, LEO satellites must transmit with sufficient power. LEO satellites also need to maintain connectivity with these UEs as they move along their orbital paths [36]. Balancing this requirement with the need to prevent harmful interference to other satellite systems' users poses challenges while maintaining the desired communication with the UE.

One primary concern in LEO constellations is the dynamic nature of interference. As satellites move across the sky, they may interfere with each other, particularly when operating in close proximity with shared frequencies. For the satellite-to-user service downlink, the interfering signals can comprise those from all satellites whose transmissions can be received by the targeted UEs [27]. While for MSS service LEO satellites can steer their main beams to prevent interference with ground

stations, the sidelobes of LEO satellite antennas can still produce harmful interference [37]. For D2D service, ground users may find themselves within the mainlobe or sidelobe areas of multiple satellites simultaneously, leading to time-varying interference effects [38]. As LEO satellites follow distinct orbits, this interference can be intermittent but persistent.

#### 2.4.2. Regulatory effort on interference management

Previous regulatory efforts have focused on interference from Non-Geostationary Orbit (NGSO) to GSO systems, while addressing interference between NGSO systems themselves relies heavily on coordination among satellite networks. of the ITU's Radio Regulations establishes EPFD (Equivalent Power Flux Density) limits to manage interference between GSO networks and NGSO systems [36]. Recommendation ITU-R S.672 suggests reducing the recommended near-in-side-lobe level relative to peak gain by -20 to -30 dB [38].

For NGSO to NGSO, the FCC in the US has introduced measures to encourage NGSO systems to coordinate with other NGSO systems [39]. However, current efforts are limited to information sharing without clear requirements or enforcement methods. One coordination method is 'look-aside' where at least one of the interfering systems uses a different satellite to communicate with the user terminal, different from the nominal satellite. If coordination fails, the FCC requires using 'band-splitting' where interfering systems equally divide the common frequency band. However, band-splitting significantly degrades throughput, making its practical use limited among satellite operators.

## 3. Literature review

### 3.1. NTN 5G NR - performance analysis

Research on the performance evaluation of GSO and NGSO satellite communication systems has been ongoing for both fixed and mobile satellite services. However, comprehensive performance analyses for the latest satellite direct-to-device based on 3GPP NTN standards remain relatively limited.

Research in [40,41] employs stochastic geometry to analyse downlink coverage probability and average data rate for large-scale LEO constellation networks[41]. results indicate that interference and fading channels significantly undermine coverage performance. The study by [42] evaluates system-level performance through link budget computations for both GEO and LEO systems. It highlights that without proper interference management techniques, frequency reuse can lead to severe interference, leading to most users experiencing outages. While these studies highlight mega-constellations performance, their analyses primary are toward higher frequency bands and high data rate service using VSAT as receivers, they may not reflect performance for satellite D2D service.

[43] is one of the most recent studies analysing the throughput and capacity performance of a LEO satellite-based NTN system. It reveals a downlink capacity of approximately 600 Mbps for an LEO satellite in S band with 30 MHz bandwidth catering to UE. The average downlink per user throughput supported is approximately 2 Mbps, and it will decrease with user density increase. While the study is more recent and specific to the 5G NR NTN scenario, this study primarily focuses on fixed, ground-level receivers and does not consider interference effects, which could impact real-world performance. [44] is another research targeting 5G NR NTN technology. It examines handover performance in the LEO-to-5G NR UE scenario. The results show NTN performed worse in terms of handover and radio link failures compared to terrestrial networks in scenarios such as high-speed trains, due to late handovers caused by the relative speed of cells and high downlink interference. The study underscores the challenges of maintaining seamless connectivity in high-speed and long-distance LEO-based networks.

The study in [2] explores optimising various physical layer aspects of 5G-NR, including numerologies, modulation and coding scheme (MCS) levels, and multiple-input multiple-output (MIMO) schemes to meet the characteristics of NTN satellite systems. Simulation outcomes demonstrate that NR-PDSCH achieves a throughput of 30 Mbps under 3GPP NR-NTN channel models for stationary UE when employing high MCS levels such as 256-QAM (Quadrature Amplitude Modulation), with an SCS of 15 kHz. It also demonstrates the necessity of using higher SCS of 120 kHz to accommodate high mobility scenarios, such as speeds of 500 km/h. These physical layer optimisations are shown to enhance link reliability for NTN to support high mobility scenarios.

### 3.2. NTN connectivity challenges

Recent academic studies have extensively examined the key technical challenges in satellite direct-to-device communications ranging from link budget constraints and system integration requirements to performance limitations.

Studies [45–47] analyse direct satellite-to-device (D2D) connectivity challenges. Boumard et al. [45] evaluates six satellite systems (Inmarsat, EchoStar, Iridium NEXT, GLOBALSTAR-2, Lynk, and AST SpaceMobile), finding that while most can support narrowband IoT services in downlink, uplink remains challenging due to limited device antenna gain, with only AST SpaceMobile capable of broadband connections due to its low orbit and high EIRP. Sattarzadeh [46] examines 5G NTN integration, focusing on RAN modifications, demonstrating that beam coverage must be confined to meet carrier frequency offset requirements. Pastukh et al. [47] investigates hybrid satellite systems operating within terrestrial IMT bands, revealing SINR variations from -5 to 12 dB with 1–2 dB degradation causing synchronisation failures and throughput reduction, proposing frequency-territorial separation as a mitigation strategy.

Research in [48] provides a comprehensive survey of multi-satellite MIMO systems that leverage satellite cooperation and advanced technologies like Orthogonal Time Frequency Space (OTFS) to overcome link budget challenges in D2D connectivity. Study in [49] proposes integrated terrestrial-satellite relay networks (ITSRNs) with aerial relays, focusing on maximising throughput and minimising energy through optimisation of power allocation and relay altitude within a single satellite network architecture. Pastukh et al. [50] concentrates on the critical challenge of Doppler-shift compensation for supported smartphones in NTN systems, demonstrating through simulations that services degrade by approximately 12.8 times without proper Doppler mitigation.

Although these studies provide valuable insights into specific technical challenges, they primarily focus on single-system optimisations without addressing the emerging environment of multiple competing constellations. Boumard et al. [45]'s static link budget analysis and [48]'s cooperative multi-satellite architectures do not consider inter-constellation interference, which our study demonstrates can cause severe degradation to neighbouring systems. Unlike [46]'s focus on RAN adaptations and signalling procedures or [50]'s emphasis on Doppler-shift effects, our analysis quantifies actual performance metrics for moving UEs, revealing that spectrum bandwidth constraints and inter-constellation interference, rather than protocol-level issues or Doppler alone, can fundamentally limit achievable data rates. Moreover, while [47] addresses terrestrial-to-satellite interference and [49] optimises single-network architectures using aerial-relay solutions, our research examines inter-constellation interference, practical operating thresholds and interference mitigation strategies (e.g., sidelobe control) for real-world scenarios where multiple non-cooperating satellite operators coexist. This work provides design guidelines that bridge the gap between theoretical single-operator optimisations and complex multi-constellation commercial deployments.

### 3.3. NTN Interference effect and management

Previous studies regarding interference of satellite systems focus on between NGSO and GSO and in high-frequency bands such as Ka and Ku band, published works regarding interference between NGSO systems are fewer, only a handful target the recent NTN technology.

#### 3.3.1. Interference effect

Research in [51] investigates the impact of interference from NGSO orbit links on GSO orbit links using BER performance as a metric. The study concludes that the angle between the interferer's orbit and the GSO orbit has a greater influence on system performance compared to the number of interference orbits. The study in [52] examines different large constellations in the Ku-band, such as Starlink, OneWeb, and Kepler, assessing their interference effects on the GSO system and each other's performance based on throughput. The research indicates that existing spectrum regulations and traditional interference mitigation techniques, such as look-aside and band-splitting, are not sufficient to protect GSO systems from NGSO interference or from interference among NGSO systems, especially for downlink interference. It points out that the high transmit power of LEO satellites, e.g., Kepler satellites, can lead to severe service degradation for other NGSO systems. The study concludes that the complexity of interactions in new NGSO systems requires more sophisticated engineering solutions and potentially stricter regulatory requirements.

The authors in [53] present the potential interference risks posed by Line-of-Sight (LOS) channels between terrestrial base stations and satellites, specifically impacting uplink satellite communications. The research in [54] investigates the interference effects of spectrum sharing between terrestrial and satellite systems, with its numerical findings highlighting the significant degradation of satellite link outage performance due to uncontrolled terrestrial interference.

Some of the more recent studies [55,56] examine the interference issues between different satellite operators on the newly assigned NTN frequency bands. The authors in [55] conducted interference analyses in the use of L-band and S-band frequency ranges for NTN technologies. The study examined the coexistence of NTN satellite systems with existing MSS systems in the two bands. By using STK software to simulate NTN satellite constellations, the research found Signal to Noise Ratio (SNR) degradation up to 6 to 8 dB across the current MSS networks. On the other hand, Pastukh et al. [56] focuses on interference analysis concerning Starlink Gen2 satellites operating within the 1990–1995 MHz frequency band used by T-Mobile in the United States. The research assesses potential interference from Starlink's satellites to MSS and NTN satellites in orbit operating within the n256 bands. The simulation-based evaluations reveal up to a 40 % reduction in uplink throughput and service degradation for existing MSS and future NTN satellite systems. The two research results particularly highlight severe co-channel interference effects due to the conflict of frequency bands used by different satellite systems and the potential impact of overlapping by satellite operators using the MNO spectrum in a particular country and the MSS S-band for NTN assigned by an international body.

Both recent studies in [55] and [56] explore interference between satellite systems, specifically modelling NTN. However, similar to prior studies, the interference analysis in [55] focuses on interference toward satellites and MSS Ground Stations (GS), while [56] emphasises interference between satellites in orbit, which have higher receiver gain. It is evident that there is a gap in research concerning the development of satellite-to-user device communication. The characteristics differ significantly, with GS having large antenna gain and higher G/T, while handheld devices exhibit much lower G/T.

In this study, we aim to delve into the scenario of satellite direct-to-device downlinks for moving objects where the UE possesses a low G/T ratio. We will investigate how varying interference levels may impact the user's Quality of Service (QoS).

#### 3.3.2. Interference management

Recent academic efforts for interference management have been focusing on using Machine Learning (ML) and Artificial Intelligence (AI) approaches. The papers in [57] and [1] analyse the use of cognitive satellite communication networks to mitigate the impact of interference. Liu et al. [57] focuses on employing coordinated protocol mechanisms, real-time database management, and spectrum sensing technology to enhance interference avoidance capabilities. Enabled by the Radio Environment Map (REM), [1] employs planar antenna array (PAA) beamforming techniques, with knowledge of interference locations and strengths provided by the REM, the result demonstrates that the framework can effectively mitigate interference. Authors in [37] address time-varying interference caused by overlapping main and sidelobes of satellite beams when there is a lack of direct coordination capabilities between systems. The research proposes leveraging statistical learning and deep reinforcement learning techniques to maximise throughput and manage time-varying interference. Saifaldawla et al. [58] introduces an unsupervised learning methodology grounded in the Autoencoders model (AEs) for interference detection. The numerical results demonstrate high accuracy in detecting interference in the received signal. The research in [59] outlines a framework demonstrating the application of ML across various use cases and network layers within NTN environments. A reinforcement learning (RL) algorithm is highlighted for its ability to dynamically select the most suitable transmitters, whether terrestrial gNB or NTN, in a given area.

### 3.4. Other NTN aspect - direct-to-satellite IoT

IoT services based on LEO satellites, often referred to as Internet of Remote Things (IoRT), have attracted interest from private companies and academia due to their potential to connect remote, dispersed, or inaccessible devices across wide geographical areas, enabling applications ranging from environmental monitoring to smart Agriculture.

Studies [60] and [61] both investigate LoRa-based approaches, with [60] analysing packet collision probabilities under ALOHA protocols and highlighting LR-FHSS's scalability gains. Álvarez et al. [61] proposes trajectory-based uplink transmission policies that exploit satellite orbital information. Afhamisis and Palattella [62] introduces SALSA (Scheduling Algorithm for LoRa to LEO Satellites), which replaces traditional ALOHA with a TDMA scheduler for LoRaWAN-to-LEO that tackles synchronisation and intermittent visibility, scheduling uplinks by satellite availability periods and footprints.

While [60–62] concentrate on uplink collision probabilities and energy-efficient transmission policies for IoT devices, this work examines the challenges of providing continuous service to moving User Equipment (UE) in unserved or remote areas, particularly focusing on achieving higher data rates beyond basic IoT applications. Whereas previous studies optimise for low-power, delay-tolerant IoT scenarios, this research tackles the more demanding requirements of enhanced mobile broadband (eMBB) services, demonstrating that current spectrum allocations are sufficient for messaging and voice but probably inadequate for internet browsing, particularly due to the dynamic interference challenge between satellite systems.

## 4. System model

### 4.1. Use case scenario

Given the recent occurrences of natural disasters across the world, coupled with the limitations of terrestrial network coverage and infrastructure damage, ensuring communication in unserved and remote areas has become a critical concern for emergency services and disaster relief efforts. While the deployment of VSATs in fixed locations to provide satellite MSS service has commenced in unserved and remote areas and evacuation centres [63], the need for continuous coverage for emergency personnel and the mobility offered by current smartphones in ensuring seamless communication has drawn significant attention from government organisations to the development of 5G NTN technology.

Based on this background, we choose Australia as an example. We assume the Australian government is undertaking disaster planning with the objective of evaluating the efficacy of integrating 5G NTN technology to ensure uninterrupted communication for emergency services,

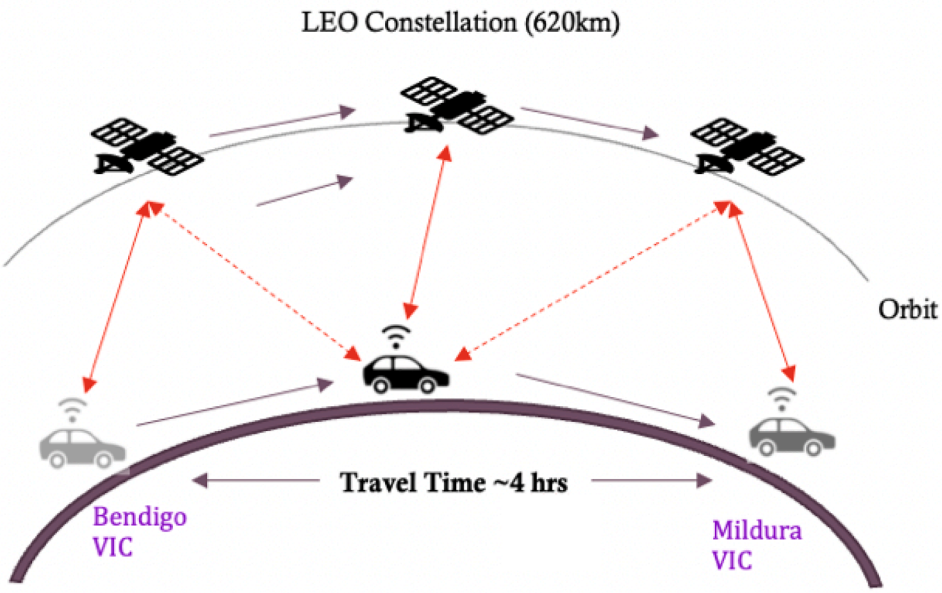



**Fig. 4.** NTN - 5G NR baseline scenario.

particularly in dynamic scenarios involving moving vehicles or personnel in disaster-prone regions. The study will focus on two scenarios: the baseline scenario, where a LEO constellation employing 3GPP NTN technology in the S-band (n256) operates without inter-satellite system interference, and the second scenario simulating NTN-to-NTN system downlink interference. To evaluate system performance and interference effects, MATLAB satellite toolbox simulations will be conducted to obtain performance metrics.

The simulated NTN system architecture features direct access to UE and a transparent payload satellite system. The LEO systems utilise walker delta and walker star constellation configurations with steerable, multibeam satellite antennas. Additionally, we assume the use of Global Navigation Satellite Systems (GNSS) for pre-measuring to compensate for Doppler-shift effects, and the satellite links operate under Line of Sight (LOS) and clear-sky conditions, neglecting shadowing and multipath effects.

The baseline scenario (Fig. 4) will analyse the performance of 5G NTN downlink D2D to a travelling UE. Assuming a travel time of approximately 4 h by vehicle between two suburban locations, Bendigo and Mildura, a scenario comprising the serving LEO constellation and the UE will be created (Fig. 5). Walkerstar constellation is used to assemble Omnispace's LEO system, with parameters derived from their deployed Spark 2 satellite and the press release detailing their plan to provide IoT/5G NTN service using 200 LEO satellites with 3GPP assigned n254 S-band [64]. The performance analysis will focus on the LEO constellation's ability to maintain LOS during the travel time and support a positive link Margin with the required minimum BER throughout the journey to ensure service quality.

For the interference scenario (Fig. 6), co-channel interference between two NTN LEO satellite systems will be assumed, with both systems using the same frequency spectrum range and providing downlink services to users. The actual minimum distances between the UE and the serving satellite, and between the UE and the interfering satellite, are 620 km and 525 km, respectively, based on the constellation's altitude (Table 8). The actual distances between the UE and both serving and interfering satellites vary continuously due to satellite movement and changing elevation angles, where 620 km and 525 km (based on constellation altitudes in Table 8) represent the minimum distances at 90°elevation angle; these dynamic distances and corresponding Receiving Signal Strength (RSS) are simulated and obtained using MATLAB's satellite toolbox. Aggregated interference will be considered to reflect the effect of large-scale constellations within the LOS of the receiver. The serving satellite system will be simulated using parameters from Omnispace's LEO system, while the interfering system will be represented by a simulation of Starlink's constellation. The Starlink constellation refers to their proposed use of 840 satellites with direct-to-device payloads as outlined in their request to test the 5G NTN service [36].

Fig. 7 presents the two simulated NTN constellations. Multibeam directional antennas will be assumed for both serving and interfering satellites, with their mainlobes directed towards serving users while their sidelobes act as interfering signals to other systems' serving users. To ensure accuracy in assessing interference effects, different Equivalent Isotropic Radiated Power (EIRP) levels will be used for serving and interfering satellites in the simulation. The sidelobe EIRP of the interfering LEO is adjusted to -10, -20, and -30 dB from the mainlobe, based on ITU-R S.672 recommendations for comparison [38].

Detailed simulation parameters for both scenarios are listed in Table 8 to evaluate system performance and interference effects. These parameters are based on the simulation guidelines provided by ITU-R M.2514 IMT-2020, with handheld (UE) specifications aligning with those outlined for rural (eMBB) and rural (eMTC) in the guidelines. A channel bandwidth of 5 MHz for 5G NTN is used as the base, assuming a minimum block of spectrum obtained for one operator.

## 5. Simulation method

A link budget model is computed to provide a theoretical / numerical result reference based on LEO satellite distance and a fixed receiver location, demonstrating the link margin for different channel bandwidth in the design.

MATLAB's Satellite toolbox [65] is used to simulate the connection between the constellation and the moving UE. The access analysis function is used to assess the LOS visibility from LEOs to the UE during the trajectory. Receiving Signal Strength (RSS) will be measured every 30 seconds during the 4 h traveling period for the baseline scenario and every 5 m for the interference scenario. The 48-beam antenna pattern of the satellite is based on the custom antenna function (HelperCustom48BeamAntenna) provided by MATLAB's Phased Array System Toolbox [66].

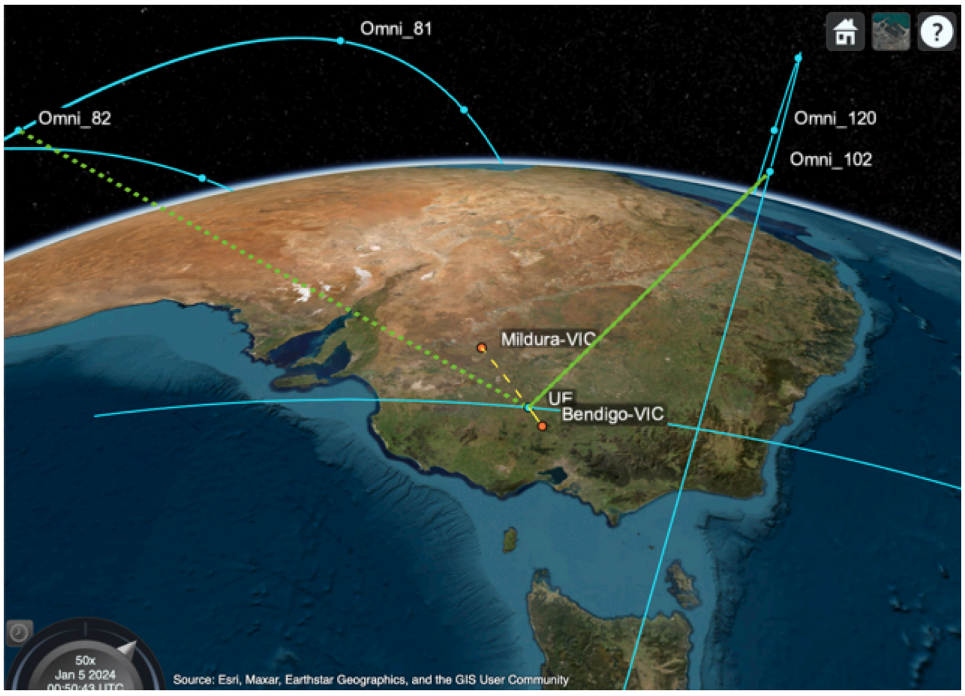



**Fig. 5.** NTN - 5G NR performance analysis simulation.

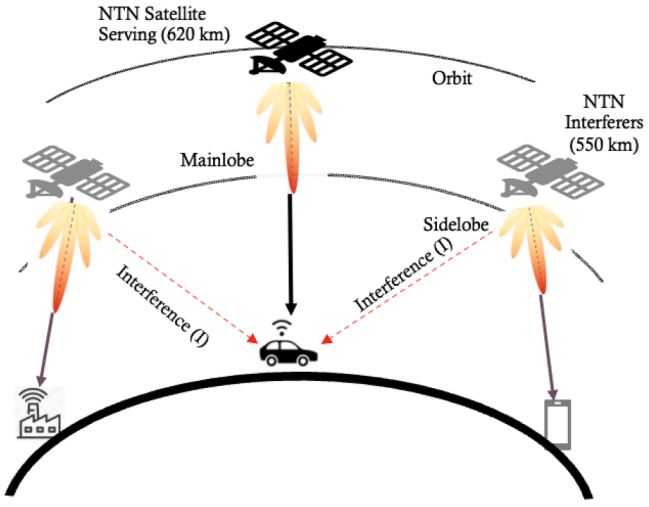



**Fig. 6.** NTN - inter-satellite system interference scenario.

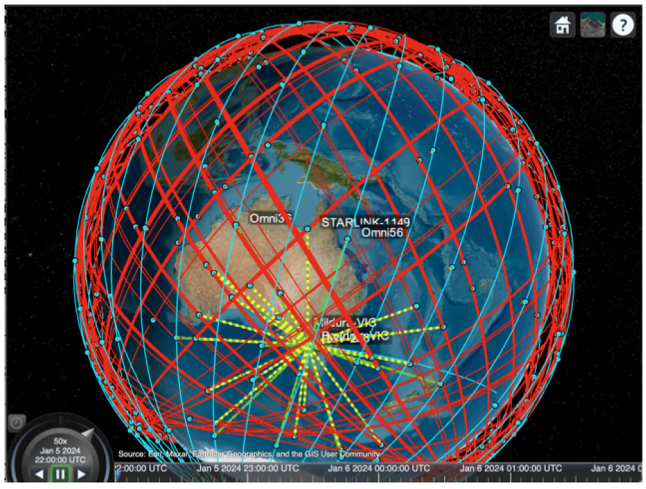



**Fig. 7.** Simulated inter-NTN systems interference.

### 5.1. QoS parameters and formulas

Effective Isotropic Radiated Power (EIRP) represents the power transmitted by the Satellite multiplied by the directional gain of the antenna. Effective Isotropic Radiated Power (EIRP) density is the signal strength spectral density that the antenna can radiate per unit bandwidth [28], EIRP will be adjusted based on the simulated Bandwidth (BW), and is obtained by:

$$\text{EIRP (dBW)} = \text{EIRP density} + 10\log_{10}(\text{BW}). \tag{1}$$

The Carrier-to-noise power ratio (CNR) is given by:

$$C/N \text{ (dB)} = \text{EIRP} + G/T - k - \text{PL} - B, \tag{2}$$

where EIRP is the Effective isotropic radiated power (dBW), $G/T$ is the antenna gain-to-noise-temperature ratio (dB), $k$ is the Boltzmann constant (-228.6 BW/K/Hz), $PL$ is the free space path loss (dB), and $B$ is channel bandwidth (dB/Hz).

The energy per bit to noise power spectral density ratio ($E_b/N_0$ )can be obtained by:

$$E_b/N_0 \text{ (dB)} = C/N_0 - 10\log_{10}(R_b), \tag{3}$$

where $C/N_0$ is the carrier-to-noise power density ratio and $R_b$ is the bit rate.

For a given bandwidth, the $E_b/N_0$ ratio has to be large enough to achieve the transmission bit rate with an acceptable error performance in terms of bit error ratio or probability of bit error. Assuming the downlink uses the common QPSK modulation scheme with no code rate consideration, and a BER threshold of $10^{-5}$ (the minimum for voice service in general), the required minimum $E_b/N_0$ energy to noise ratio to achieve a bit error probability of less than $10^{-5}$ can be calculated by:

$$P_e = 10^{-5}, \quad P_b \approx Q\left(\sqrt{2\frac{E_b}{N_0}}\right). \tag{4}$$

The required $E_b/N_0$ is assumed to be 9 dB in our simulation.

**Table 8**
Simulation parameters: LEO downlink to UE.

| Parameters | Serving | Interferer |
|---|---|---|
| Frequency (Hz) | 2.1GHz (S band) | 2.1GHz |
| Bandwidth (Hz) | 5–25 MHz – (5G) | 5–25 MHz – (5G) |
| **Satellite Transmitter** | | |
| Satellite System | LEO | LEO |
| Type of orbit | Circular | Circular |
| Constellation model | Walker Star | Walker Delta |
| Altitude (km) | 620 (Omnispace) | 525 (Starlink) |
| No. of orbital planes | 10 | 14 |
| No. of satellites per plane | 20 | 60 |
| Total No. of satellites | 200 | 840 |
| Inclination Degree | 97.4 | 53 |
| Antenna | Multibeam | Multibeam |
| EIRP density (dBW/MHz) | 34 (Mainlobe) | 34 (Mainlobe) |
| Max antenna Gain (dBi) | 30 | 30 |
| Propagation | Line of sight (LOS), Clear sky | Line of sight (LOS), Clear sky |
| **Receiver (Handheld / UE)** | | |
| UE Height (m) | 1.5 | 1.5 |
| Antenna Type | Omni-directional antennas | Omni-directional antennas |
| G/T - User (dB/K) | -31.6 | -31.6 |
| Noise Temp (K) | 290K | |
| Noise Figure (dB) | 7 dB | |
| Modulation | QPSK | |
| Required Eb/N (dB) | 9 dB | |

The corresponding link margin from the satellites to the UE (downlink) is obtained by:

$$M(\text{dB}) = E_b/N_0 \text{ (received)} - E_b/N_0 \text{ (required)}. \quad (5)$$

In our Link Budget table, we add a case scenario that Forward Error Correction (FEC) coding is used for reference. FEC provides coding gain that usually resulting in improvement in link performance. The formula for $E_b/N_0$ ratio with code rate consideration becomes:

$$E_b/N_0 \text{ (dB)} = C/N_0 - 10\log_{10}(R_b) - 10\log_{10}(1/r), \quad (6)$$

where $r$ is the code rate (e.g., rate -3/4 turbo code has $r = 0.75$). The term $10\log_{10}(1/r)$ represents the rate penalty in dB, which accounts for the additional bandwidth required due to coding redundancy.

For link-level performance for 5G NR NTN, 3GPP has introduced a link adaptation method in Release 18. The throughput can be approximated by an attenuated and truncated form of the Shannon bound [67]. Given a specific SINR measured in dB, for $\text{SINR}_{\text{min}}$ and $\text{SINR}_{\text{max}}$ between -10 and 30 dB, the adapted Shannon bound formular is [67]:

$$\text{Throughput (SINR) bps/Hz} = \alpha \cdot S(\text{SINR}), \quad (7)$$

where $\alpha$ is the attenuation factor (0.6 for downlink), and

$$S(\text{SINR}) = \log_2(1 + \text{SINR})\text{bps/Hz}.$$

The Carrier-to-Interference and Noise Ratio (CINR) interference metric is used to determine the level of interference caused by the LEO satellite on another LEO system's Downlink user. We assume the serving UE receiver is subjected to aggregated interference signal power from all the interfering system's satellites on the same channel, calculated as

$$I_{\text{agg}} = \sum_{i \in \{\text{interferers}\}} I_i. \quad (8)$$

The CINR is the ratio between the Receiving Signal Strength (RSS) in Watt and the interference power $I_{\text{agg}}$, plus the noise power (N),

$$\text{CINR} = \frac{\text{RSS}}{I + N}. \quad (9)$$

## 6. Results

In this section, we present the baseline link budget table for three different bandwidths (5 MHz, 10 MHz, and 25 MHz) as a reference, and the simulation results obtained in both scenarios using a 5 MHz bandwidth to evaluate performance.

We utilise cumulative distribution functions (CDF) to illustrate various performance metrics (CNR, CINR and Throughput) for downlink service performance. For the interference scenario, CDF is used to compare the reduction in CINR and throughput loss. According to ITU Recommendation ITU-R S.2131 for satellite systems typically C/N reduction of 1 dB is corresponds to 10 % reduction in spectral efficiency [68], which is considered tolerable signalling degradation [55]. In 3GPP Release 18, for NTN performance metric measurement, the average throughput loss should be less than 5 % [67].

We also include scenarios with Forward Error Correction (FEC) consideration in the link budget results to demonstrate the potential performance gains with turbo coding. However, our simulations will focus on the performance of dynamic LEO constellation systems and moving objects, analysing the varying interference effects without FEC implementation. We use this worst-case scenario approach because, while link budgets represent static, point-to-point scenarios where FEC gains are predictable and well-documented, our dynamic simulation involves time- and position-varying interference patterns and constellation dynamics that affect the signalling channel before error correction is applied. By separating these analyses, we can clearly identify the contribution of each system component—the Additive White Gaussian Noise (AWGN) channel conditions, interference scenarios, and the coding gains—providing a more comprehensive analysis of the performance and challenges posed by the coexistence of multiple LEO constellations.

### 6.1. Link budget for different bandwidths

Table 9 illustrates the satellite direct-to-UE link budget with different channel bandwidths of 5, 10, and 25 MHz. To achieve a BER of $10^{-5}$ using QPSK modulation, the optimal bit rates of 1, 3, and 6 Mbps are shown, with a link margin of 5 dB for the uncoded scenario.

The link budget also considers Forward Error Correction (FEC) using turbo codes. Turbo code is an error correction technique that employs two parallel recursive systematic convolutional (RSC) encoders separated by an interleaver and uses an iterative A Posteriori Probability (APP) decoder [69]. It is a widely used forward error correction scheme for satellite and wireless communications due to its ability to achieve exceptionally low bit error rates (BER) at signal-to-noise ratios (SNR) very close to Shannon's theoretical limit [69].

For the scenario with FEC, assuming 3/4 turbo code, the $E_b/N_0$ is obtained using formula (6). The required $E_b/N_0$ is assumed to be approximately 3.5 dB [69,70] for achieving a BER of $10^{-5}$ in our link budget calculation. As shown in Table 9, the link margin improves from 5 dB without FEC to approximately 9.3–9.5 dB with 3/4 turbo coding across all bandwidth configurations, providing a coding gain of about 4.3–4.5 dB. While the $E_b/N_0$ decreases by 1.25 dB due to rate penalty (e.g., bandwidth overhead from redundant bits, reduced spectral efficiency), the turbo coding scheme provides coding gain that exceeds the rate penalty, resulting in improvement in link performance.

The link budget provides a baseline based on the fixed distance between the satellite and the UE using an EIRP density ratio of 34 dBW/MHz per ITU-R M.2514 IMT-2020. It shows that, for D2D services, the link budget required to achieve higher bit rates is largely constrained by the low G/T of the handheld UE. It also indicates that, to increase the data rate, a larger spectrum bandwidth is needed. However, due to the scarcity of spectrum, another industry approach is to increase the EIRP, which may heighten the risk of interference.

### 6.2. LEO Constellation downlink to 5G NR UE performance results

#### 6.2.1. Baseline scenario without interference

Providing continuous coverage is one of the main objectives of using LEO constellations in NTN. Fig. 8 shows that the UE remains within

**Table 9**
Link budget - satellite to UE.

| Link Budget | | DL in different BW | | |
|---|---|---|---|---|
| Parameters | Unit | Value | Value | Value |
| Downlink Frequency | GHz | 2 | 2 | 2 |
| Bandwidth | MHz | 5 | 15 | 25 |
| Distance | km | 620 | 620 | 620 |
| Free space pathloss | dB | 155 | 155 | 155 |
| Other losses[1] | dB | 8 | 8 | 8 |
| Data rate (QPSK) | Mbps | 1 | 3 | 5 |
| **Satellite Transmitter** | | | | |
| EIRP Density | dBW/MHz | 34 | 34 | 34 |
| EIRP | dBW | 40 | 45 | 47 |
| **UE Receiver** | | | | |
| Receiver Gain | dBi | 0 | 0 | 0 |
| G/T | dB/K | −31.6 | −31.6 | −31.6 |
| **Noise Assumption** | | | | |
| Boltzmann's constant | dBW/K/Hz | -228.6 | −228.6 | −228.6 |
| Noise Temp | K | 290 | 290 | 290 |
| Noise Figure | dB | 7 | 7 | 7 |
| Noise power density ratio | dB/Hz | −197 | −197 | −197 |
| Noise power | dBW | -130 | -125 | -123 |
| **Performance - Uncoded** | | | | |
| Received Power | dBW | −123 | −118 | −116 |
| CNR[2] | dB | 6.99 | 7.22 | 7.00 |
| $C/N_0$[3] | dB | 74.0 | 79.0 | 81.0 |
| $E_b/N_0$[4] | dB | 14.0 | 14.2 | 14.0 |
| Req. Eb/N0 (-$10^{-5}$ BER) | dB | 9 | 9 | 9 |
| Link Margin | dB | 5 | 5 | 5 |
| **$E_b/N_0$ - with FEC** | | | | |
| $E_b/N_0$ | dB | 12.8 | 13 | 12.8 |
| Req. Eb/N0 [5] | dB | 3.5 | 3.5 | 3.5 |
| Link Margin | dB | 9.3 | 9.5 | 9.3 |

[1] Includes atmospheric absorption, rain fade, and implementation losses.
[2] Carrier to Noise Power Ratio
[3] Carrier to Noise density Ratio
[4] Energy bit per noise density
[5] Assumption with turbo 3/4 code rate for (-$10^{-5}$ BER) [69,70]

the line of sight (LOS) of at least one LEO satellite during the 4-h period. On average, each satellite maintains access to the moving UE for approximately 240 seconds per pass.

Fig. 9 presents the simulation of 5G NTN with a channel bandwidth of 5 MHz and a bit rate of 400 kbps. The link margin results are based on the required $E_b/N_0 = 9$ dB, obtained using calculation (4) with QPSK modulation achieving a BER of $10^{-5}$ during the traveling period with sampling every 30 seconds. While the link margin is kept above 0 dB, it illustrates fluctuations during the period due to the dynamic movement of the LEO constellation and the moving UE at different pointing angles. Compared to the link budget result, the simulation indicates a lower achievable data rate due to the variation in path loss, which more closely reflects real-world scenarios.

Fig. 10 shows the CDF graph for different bit rate simulations, indicating that a bit rate of 500 kbps or less can be sustained during the entire journey (with Link Margin of 1dB or higher), while 600 kbps or above may experience periodic degradation/outages during the traveling period.

### 6.2.2. *Interference scenario*

For the interference scenario, previous studies showed simulation results where the interfering satellite's radiated power used mainlobe figures, resulting in the worst-case scenario. However, this may not reflect the real situation, where most of the new-generation satellites are capable of controlling the sidelobe power. In this study, we adjust the

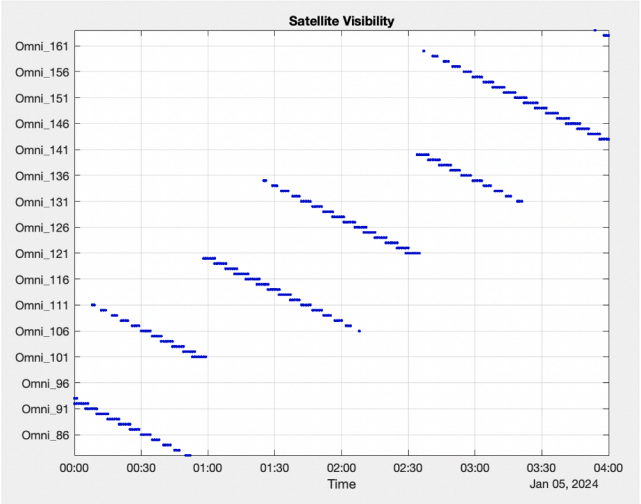



**Fig. 8.** Satellite visibility toward moving UE.

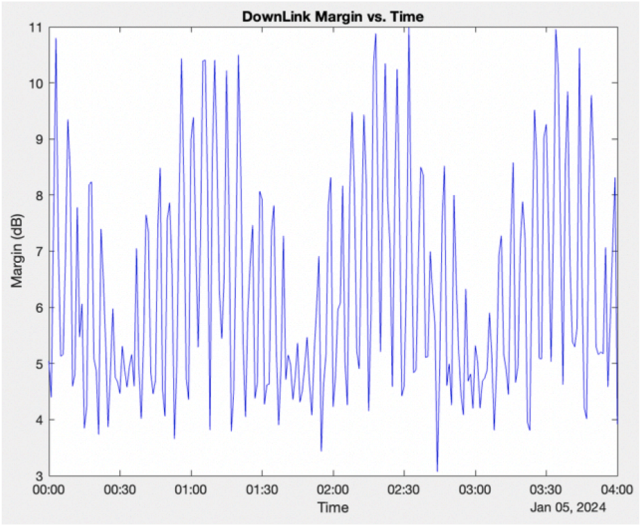



**Fig. 9.** Link margin (QPSK with bitrate = 400 kbps).

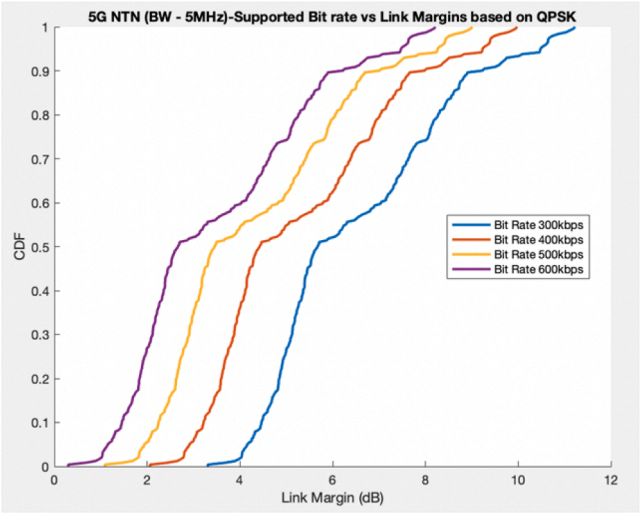



**Fig. 10.** CDF of link margins for different bit rate.

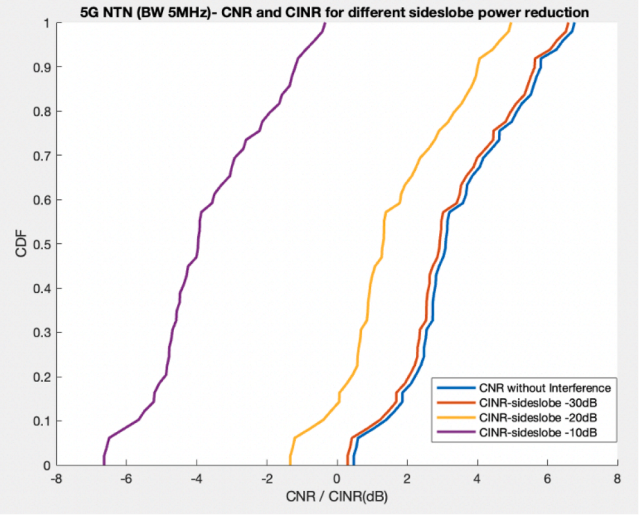



**Fig. 11.** CDF- CNR/CINR with different sidelobe power.

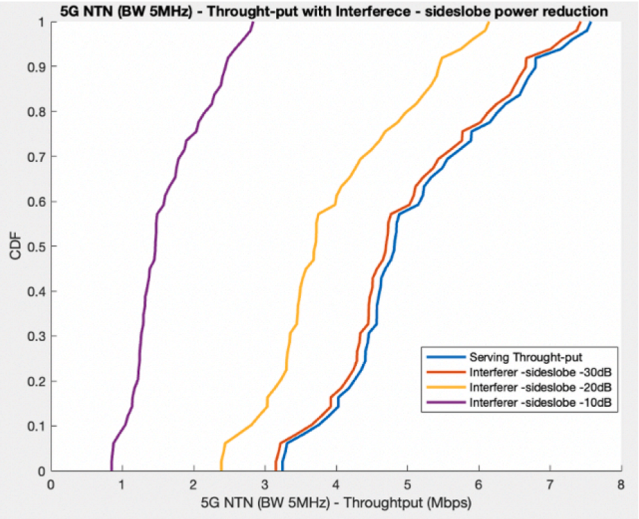



**Fig. 12.** CDF- Throughput with different sidelobe power.

EIRP of the interferers for comparison, assuming interference is aggregated from the interfering satellite system.

Fig. 11 indicates the CDF graph of CINR results comparison for CNR without interference, and with the interferer's sidelobe reduced by 10, 20, and 30 dB respectively. The results show a reduction of 8 dB for 90 % of the time when the interferers' sidelobe EIRP is -10 dB relative to the serving constellation. Only when the interferers' EIRP is reduced by 30 dB does the deviation decrease to approximately 1 dB.

Fig. 12 shows a CDF graph illustrating the throughput rate comparison for different EIRP reduction levels. The results indicate that the downlink would experience over 20 % and 70 % throughput loss with 20 dB and 10 dB EIRP reductions, significantly higher than the 5 % performance metric set by 3GPP for 5G NR [66].

The results reveal that high EIRP from interfering constellations can lead to a significant decrease in spectral efficiency and throughput loss. The link margin results (Fig. 13) also show that in the worst-case scenario, the UE will not be able to maintain acceptable BER (with 500 kbps data rate) during the traveling period. It highlights the significant effect of the interfering system reducing its sidelobe EIRP to the recommended -30 dB level to minimise harmful interference to another NTN system's users.

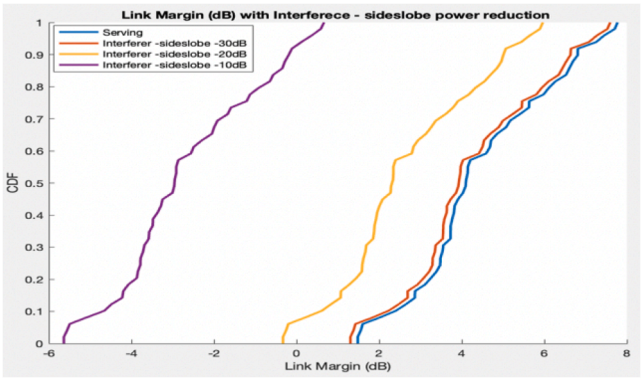



**Fig. 13.** CDF- Link Margin (500kbps) with Interference.

## 7. Analysis and discussion

The performance results for 5G NR direct satellite-to-device (D2D) communication reveal that LEO constellation can provide continuous coverage for a moving UE in unserved and remote areas, adequately supporting low data rate mobile services such as emergency alert and SMS. However, the characteristics of handheld UE with low G/T necessitate higher EIRP to compensate for the link budget and support higher data rate traffic.

The results also highlight that the current limited spectrum bandwidth poses challenges in achieving higher data rates and maintaining QoS, particularly for moving UEs. Our simulations indicate that, assuming a carrier bandwidth of 5 MHz per operator and considering the fluctuation in link margin due to the dynamic variation caused by constantly moving transmitters and receivers, the achievable bit rate for one UE throughout the traveling period is approximately 500 kbps or less, assuming an acceptable BER using QPSK modulation. If we assume 32 kbps is needed for basic messaging and voice services, the system can support approximately 10 + users simultaneously. To support higher data rate services such as internet browsing or more users, operators will face challenges in maintaining QoS for moving objects with dynamic changes in path loss.

In addition, based on the 5G NR specification and previous study's findings, to integrate NTN with 5G-NR, physical layer optimisations such as smaller cell size and larger SCS are preferred to compensate for latency and Doppler-shift of the satellite constellation and moving devices. This, however, may reduce the number of supported users per cell/beam. For a 5 MHz downlink channel bandwidth, the supported SCS is only 15 kHz; channel bandwidths of at least 10 MHz and 20 MHz are necessary to support SCS of 30 kHz and 60 kHz, respectively. This indicates that, based on the current restricted spectrum, the choices for accommodating higher SCS are limited. For NTN operators to leverage the flexibility of the 5G NR specification, larger spectrum bandwidth will be needed.

The simulation results illustrate, assuming a 5 MHz bandwidth allocation per operator, 5G NTN services toward UEs that are satellite service enabled (e.g., Samsung Galaxy S25, Apple iPhone 14 or above) or dedicated satellite UEs are sufficient for emergency services, messaging, and voice services, but not for high data rate (eMBB) services. Achieving higher EIRP and additional spectrum bandwidth will be essential to enhance the data rate and user coverage. While 3GPP recently allocated up to 80 MHz in the lower band for 5G NTN, in Australia, for example, only 25 MHz is proposed at this stage. Spectrum sharing between TN and NTN has been studied in the industry to leverage and maximise the usage of the scarce spectrum in both MNO and MSS providers. Technology enhancements on satellite transmitters and antennas have been ongoing to increase the EIRP to achieve the required performance metrics. However, in both solutions it also leads to high risk of interference.

The Interference simulation scenario highlights the need for reducing sidelobe power to mitigate interference effectively. In reference to earlier section, both Starlink and AST SpaceMobile conducted tests demonstrating direct-to-device downlink data rates exceeding 10 Mbps using current MNO's spectrum. It is noteworthy that Starlink's tests utilised their maximum transmit EIRP of 58 dBW with a maximum allowed PFD of -80 dBW/m2/MHz [28]. Even assuming reduction in its sidelobe power by -30 dB, Starlink's sidelobe power of 28 dBW is approximately equal to our interferer system with a -10 dB sidelobe reduction in our interference simulation. Our results indicate this would lead to an 8 dB reduction in CINR and over a 70 % throughput loss for the serving NTN system's users. The results illustrate how different maximum EIRP levels for different constellations affect another satellite system's downlink performance. Given the low G/T of the UE and the race to achieve higher data rates, the situation is expected to worsen for D2D services, especially for smaller satellite operators with smaller constellations and antennas, without proper regulation and technological advancements.

The performance results demonstrate that achieving higher data rates through increased EIRP requires implementation of sidelobe power control to mitigate interference toward neighbouring satellite systems. Our analysis establishes quantitative operating thresholds for 5G NR NTN downlinks to moving UEs, specifying the minimum sidelobe reduction required at various EIRP levels under current spectrum constraints. As this study highlights the complex trade-offs between dynamic channel conditions, interference patterns, and achievable data rates/throughput, detailed design and simulation for moving-object scenarios are essential for accurate performance evaluation, rather than relying solely on traditional one-dimensional link budget analysis. The simulation framework can also be applied to other mobility use cases, such as autonomous vehicles and tracking vessels. These findings provide practical design guidelines, emphasising the need for downlink-centric mitigation technologies on the satellite side, such as beamforming [1] and beam-hopping [2], to increase mainlobe EIRP while dynamically controlling sidelobes to limit cross-constellation interference.

## 8. Future work

In future work, the integration of NTN for future 6G networks will continue to evolve, with a focus on addressing key challenges and leveraging advanced technologies to enhance performance. As highlighted in this paper, there exists a trade-off between increasing EIRP to meet higher data rate demands within a limited spectrum and the risk of harmful interference between non-coordinated satellite systems.

To address this challenge, advancements in LEO satellite capabilities are imperative. Industry efforts are underway, focusing on using advanced antenna technologies such as phased arrays and beam-hopping. These technologies aim to close the link budget gap and offer data rates up to the Mbps range while providing dynamic modification of transmit power to mitigate interference. In this work, our link budget and simulations are based on the same EIRP density ratio for the satellite transmitter and the same receiver G/T. In future work, we can further examine how variations in different vendors' technologies may improve transmitter and receiver losses, ultimately bridging the link budget gap.

For 5G NR direct-to-handheld services, spectrum sharing between TN and NTN will be crucial to enhancing performance and data rate, particularly in the context of limited spectrum availability in lower frequency ranges. However, managing the pairing mechanism between uplink and downlink is essential to minimise co-channel interference, especially for uplink services [53].

Moreover, machine learning (ML) and artificial intelligence (AI) approaches hold promise for addressing interference challenges. Academic studies have proposed ML/AI algorithms for dynamic interference detection and classification. These algorithms can be leveraged for predictive channel estimation to detect interference before it occurs and adjust transmission power at precise times to avoid interference, thus improving overall system performance [58].

In future research, exploring these advanced technologies and developing novel approaches for interference mitigation and spectrum management will be crucial for realising the full potential of NTN integration in future 6G networks.

## 9. Conclusion

Recent collaborations between satellite operators, mobile operators, and network and UE vendors have demonstrated the potential of direct satellite-to-device technology to achieve higher data rates than emergency alert or SMS services. These trial tests showcase the competitive landscape and the potential for interference issues without proper regulation and technological advancements. This paper addresses the performance constraints of 5G NTN direct satellite-to-device services due to the limited available spectrum and the potential risk of co-channel interference.

Our analysis provides critical insights besides increasing SINR. First, by using simulations, we quantify the trade-off for EIRP and interference, while maximising EIRP to 58 dBW can enable data rates exceeding 10 Mbps (as demonstrated by the Starlink in their trial test), it could simultaneously create interference where neighbouring systems experience up to 8 dB CINR degradation and 70 % throughput loss, establishing an estimated quantitative boundary for sustainable coexistence. Second, we demonstrate that the projected 5 MHz spectrum allocation per operator, based on a real country's proposal, may limit subcarrier spacing to 15 kHz, preventing the use of higher SCS (30–60 kHz) needed to compensate for Doppler effects in moving UE scenarios, thereby affecting achievable data rates and user experience. Third, our multi-dimensional analysis, using link budget as baseline and simulations for real-world use cases, reveals that traditional link budget calculations may not capture the dynamic interference patterns in LEO constellations with moving UEs, leading to performance predictions that may not materialise in real deployments.

Our findings emphasise the importance of ongoing technological advancements and regulatory measures to manage interference and optimise spectrum use. They highlight the need for continued innovation and collaboration among stakeholders to realise the vision of seamless, global connectivity in 6G. Future work will explore the use of machine learning and artificial intelligence to address interference challenges and enhance system performance.

## CRediT authorship contribution statement

**Oi Shan Wong:** Writing – original draft, Validation, Software, Methodology, Formal analysis, Data curation; **Mark A. Gregory:** Writing – review & editing, Supervision, Resources, Project administration; **Shuo Li:** Writing – review & editing, Supervision.

## Data availability

Data will be made available on request.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

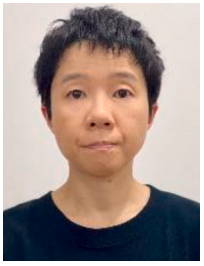
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